

Move 37: Accelerating and enhancing research with AI-powered digital assistants

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With Google Cloud, Move 37 is running AI-powered digital assistants to conduct thematic analysis, enabling researchers to reduce the time and resources needed to undertake projects by 80%.

MOVE 37

About Move 37

Founded in 2017, Move 37 is a machine intelligence company that develops algorithmic tools for critical thinking. Its vision is to power a new work operating system with augmented intelligence applications that forge a closer relationship between humans and AI.

Industries: Technology

Location: Australia

Google Cloud results

- Ensures that digital assistants can analyze tens of thousands of documents and return responses in minutes
- Enables business to run compute-intensive, leading edge AI services with just four people
- Allows business to explore development of AI-powered services that proactively act on individuals'

Reduced
80% in
project
with AI
assist

Google Cloud (<https://cloud.google.com/>)

Kubernetes Engine

Cloud Build (<https://cloud.google.com/build/>),

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Cloud GPUs (<https://cloud.google.com/gpu/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

behalf at
work

Creativity abounds in advertising campaigns, with partnerships between copywriters and art directors that can define brands for generations of customers. As a partner in an Australian advertising agency himself, Dave King nurtured and respected these relationships. In 2017, as Dave was considering how AI could deliver more relevant, personalized advertising campaigns, he began to look into how AI could replicate these relationships for knowledge workers.

"Imagine a digital assistant that deeply understands the subject or domain you work in, your history, your preference and everything else about the way you work," explains Dave. "That's the vision that prompted us to set up [Move 37](https://www.move37.ai/)

(<https://www.move37.ai/>) in 2017 and remains what we're trying to achieve."

Helping knowledge workers discover new ideas

Headquartered in Melbourne, Australia, and operating with a tight knit team, Move 37 describes itself as a machine intelligence company that develops algorithmic tools for critical thinking. Its core product is Archer, an AI-powered research assistant running on a Google Cloud infrastructure. Archer and other AI research assistants enable clients to upload documents, PDFs, research papers,

podcast transcripts and other materials for thematic analysis and summary.

"We aim to help knowledge workers connect the dots, discover new ideas and uncover insights within language," says Dave. "Our customers include analysts, strategists, consultants, researchers and marketers."

The business views collaboration between people and AI as a new field in research, and creative and critical thinking. "Humans have experience, expertise, instinct and empathy and can combine this with analysis at speed, coordination of concepts, associations between concepts, and perspectives derived from concepts by sharing these with tools like Archer," says Dave. "We're seeing a gap widen between the growing volume of data, information and content that could help knowledge workers and the time and resources these workers have to read, understand and analyze it. In harder economic times in particular, these workers are under more pressure to produce insights, analysis and thought leadership quickly and inexpensively. With Archer and other AI-powered digital assistants, we want to fill that knowledge gap."

Delivering technology, resources and support

To develop and run Archer and other digital assistants, Move 37 needed a highly scalable, dynamic, easy-to-use infrastructure, innovative AI and machine learning services, and additional

resources and expertise. After opting initially to run in other cloud services, the business elected to participate in the Google for Startups program to access support, networking opportunities and access to Google Cloud credits. "Google Cloud provided the best combination of technology, resources and support to help us achieve our business goals, and we decided to switch," says Pan Demosthenous, co-founder and Chief Technology Officer, Move 37.

Move 37 needed a considerable volume of compute power to process its AI language model in the cloud. Initially, the business wanted to consolidate its services into a containerized application environment using Kubernetes container orchestration. Using [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (<https://cloud.google.com/kubernetes-engine>) with [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>) instances running Kubernetes, Pan's team was able to easily set up development and production clusters, thus incrementally migrating components across to the cloud service during implementation. "Throughout the implementation and deployment process, we received excellent support and guidance from our dedicated account manager and Google Cloud support," says Pan. "This helped us resolve any issues and ensured a smooth transition to our new system."

Move 37 relies heavily on high-performance [Cloud GPUs](https://cloud.google.com/gpu) (<https://cloud.google.com/gpu>) to perform accelerated inference tasks. "For the first phase of the migration, we were able to quickly deploy our inference services to Google Kubernetes Engine,"

says Pan. "Google Kubernetes Engine cluster nodes support GPU drivers and NVIDIA GPU device plug-ins, which makes it easy for us to deploy and manage our GPU workloads without having to worry about the underlying infrastructure."

The business is now migrating its inference tasks to Vertex AI (<https://cloud.google.com/vertex-ai>) to take advantage of features such as automated scaling, model versioning and monitoring and logging capabilities.

Google Kubernetes Engine continues to run the core of the company's processing pipeline, while Cloud Storage (<https://cloud.google.com/storage>) stores data and content in a highly stable environment that provides easy access. This content includes build artefacts developed within Cloud Build (<https://cloud.google.com/build>) and backups, log files and other business-critical data assets.

Providing an explorable summary with rich context

With Archer running on Google Cloud, Move 37 enables researchers to work in partnership with AI to deliver thoroughly researched, relevant, and insightful projects. For example, a sustainability researcher working on changing behaviors to increase the proportion of fully electrified homes has used Archer for research, such as global measures taken to achieve this outcome. "This researcher created a short brief that Archer used to generate a preliminary research plan, which she then amended," says Dave. "Based on that review, Archer

created new research plans the researcher could amend until she was satisfied. Once she pressed submit, Archer would then discover, index and analyze at phrase-level a range of sources to pull in content.

"Within minutes, the researcher could view an explorable summary incorporating rich concepts that may, for example, comprise 100,000 words across 50 documents. Thanks to our algorithms, she could see at a glance the key themes emerging from that subject matter. She could ask questions and search to further identify the strongest concepts and the associations between them."

Archer has enabled the researcher to identify an association between people hesitant to adopt electric vehicles due to preconceptions about their range, and those reluctant to electrify their homes. "For some people, getting into an electric car is the best thing they can do to electrify their lives," says Dave. "This was an explorable, explainable concept the researcher could interrogate in a number of ways."

In another use case, an education department worker is using Archer to analyze reports into school performance, and is saving about 80 percent of the time otherwise needed to complete the exercise, according to Dave.

A third example showcases the usefulness of Archer in cases in which researchers do not have a hypothesis to test. "Two team members at a large global non-government organization have to analyze

data provided by more than 200 field workers across Australia, who engage with remote communities," says Dave. "With Archer, these analysts have cut the time to complete this analysis significantly, who could then use this time to work on different tasks."

Finally, an education consultancy is working with Archer to develop new University courses. Rather than start with a specific idea or thesis, its consultants use Archer to deliver thematic analyses based on the information available, and develop each course from there.

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Officer, Move 37*

Performing a large number of jobs concurrently

With Google Cloud, Move 37 can manage a large number of different Archer jobs concurrently without compromising performance or availability. "Until about six months ago, many people did not have a firm view about how it would feel to use an AI product," says Dave. "Now, people are increasingly being exposed to AI, and speed of compute and inference are becoming super important. At the same time, the relevance of insight, accuracy of summaries and usefulness of highlights are crucial. Google Cloud enables us to continuously improve

Archer across these measures to improve the experience for researchers in any field."

Moreover, Move 37 can continue to scale at its own pace through Google Cloud. "I've worked in the industry a long time, and when to invest in technology at scale has always been a challenge," says Dave. "Google Cloud allows us to scale as we need to and receive skilled technical support in infrastructure, and engineering along the way. Because we're doing groundbreaking work, Google Cloud occasionally had to dig deeper to help us overcome some challenges, and they've done so in a very timely way."

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In addition, with the depth of functionality delivered by Google Cloud, as well as its managed services and ease of use, the business can maintain an extremely lean workforce, similar to the model adopted by many startups providing AI products as a service. "With an eventual workforce of 15 to 20 people, we believe we can achieve extraordinary things," says Dave.

Move 37 is now looking to expand its scope beyond research, so it can help people make informed decisions and take the necessary steps to realize their goals. "We think in the immediate future, agents, research assistants, and AI thought partners will act proactively on people's behalf at work," concludes Dave. "Over time, they will get better at

understanding the way you like to think and work.

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